Recommended Assessment

Radar

Explore Radar Envelope Mode

1. Use the following table to record your observations regarding peaks of different objects, with and without the aluminum cone. Why is the peak amplitude different for different materials?

2. What are the estimated FOVs with and without the aluminum cone? How does the cone improve the reflected amplitude?

3. When you moved the reflective object farther from the radar, how did the peak amplitude change? Why?

4. When the cardboard was vertical, there was slight a reduction in peak amplitude, the metal bottle still showed significant reflection. However, at certain tilt angle, the peak of my metal water bottle significantly dropped, why is that?

Explore Radar IQ Mode

1. Write the mathematical relationship between I, Q, amplitude, and phase.

2. What does the exponential smoothing filter do to the signal, and what is the effect of decreasing α?

3. Why might engineers choose IQ mode instead of envelope mode for some applications?

Presence Detection

1. If your threshold is set too high, what type of error occurs? What happens if it is too low?

2. Suggest one improvement to the simple presence detector.

Radar Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	